
Group Project Deliverable 2

Problem Analysis and Design

Overview

In this deliverable, you will be working in your group to analyze your project idea and come up with a design for the program using an Object-Oriented Programming (OOP) approach.

Requirements

In this assignment you will need to submit a report that contains the following items. Prepare your report in PDF format.

1. Problem Analysis

As for this task, you are expected to conduct a **group discussion or brainstorming session** to analyse the problem of your project using an OOP approach. In this regard, you will be basing your analysis on the storyboard you did in Deliverable 1. You should record your session. You may want to conduct this task online for easy recording.

Things to do in this task:

- a. Identify the objects and then the classes involved in the project. As for the class members (i.e., attributes and methods), include only the significant ones.
- b. Identify class relationships. Here, analyze two things:
 - which classes that involve in any association relationship, and
 - which classes that involve inheritance relationships.

For each relationship above, justify your answer. For example, why did you think that the classes are in those kinds of relationships, or in other perspective why those classes should be modeled with those relationships.

2. Design

In this section you will document your findings from the analysis process in the previous task into an UML class diagram. The class diagram should include:

- a. Classes – including attributes and methods (enough only with the significant ones)
- b. Class relationship – including association and inheritance.

Note: Use a diagram tool such as draw.io to draw the class diagram.

Deliverables Items

There are THREE (3) items to be submitted for this deliverable:

- a. Video: group discussion or brainstorming session for the problem analysis. Host the video on YouTube. You do not have to edit the video. Submit as is. (group submission).
- b. Report about the design (group submission).
- c. Peer and Self Assessment (individual submission).

Assessments

This deliverable carries **6%** weightage for the final grade of this course. The breakdown weightage is as follows:

Criteria	Weightage
1. Problem Analysis (from the brainstorming video) - Identify objects and classes. - Identify relationships	1% 1%
Design in Class Diagram - Classes - Relationship	2% 1%
Peer and Self-Assessment 2 (PSA2) This PSA is an individual submission. Should you fail to submit, a zero mark will be given automatically.	1%
Total	6%

Submission

- Deadline: as per stated in eLearning.
- Late submission: 20% penalty for every 1 day late.
- Group Submission (for the report) :
 - Only one member from each group needs to do this submission.
 - Submit the report to eLearning.
 - Upload the video on YouTube and submit only the link to eLearning.
- Individual Submission (for Team Working):
 - submit the Peer and Self-Assessment form to eLearning.

Resources

- Resources about OOP: Lecture Topic 1 – OOP principles and UML
- Briefing on Problem Analysis and Design (for Deliverable 2) – from the class past semesters

<https://utm.webex.com/utm/ldr.php?RCID=c7f854d059ccb80bf14387a834f3705c>

Password: HelloWorld2022